

1. The value of $\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ$ is _____.
[2023 (06 Apr Shift 2)]
2. The value of $36(4 \cos^2 9^\circ - 1)(4 \cos^2 27^\circ - 1)(4 \cos^2 81^\circ - 1)(4 \cos^2 243^\circ - 1)$ is
[2023 (08 Apr Shift 2)]
 - (1) 54
 - (2) 18
 - (3) 27
 - (4) 36
3. $96 \cos \frac{\pi}{33} \cos \frac{2\pi}{33} \cos \frac{4\pi}{33} \cos \frac{8\pi}{33} \cos \frac{16\pi}{33}$ is equal to
[2023 (10 Apr Shift 1)]
 - (1) 3
 - (2) 1
 - (3) 4
 - (4) 2

ANSWER KEYS

1. (4)

2. (4)

3. (1)

1. (4)

Given,

$$\begin{aligned} & \tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ \\ &= (\cot 81^\circ + \tan 81^\circ) - (\tan 27^\circ + \cot 27^\circ) \\ &= (\tan 9^\circ + \cot 9^\circ) - (\tan 27^\circ + \cot 27^\circ) \\ &= \left(\frac{\sin 9^\circ}{\cos 9^\circ} + \frac{\cos 9^\circ}{\sin 9^\circ} \right) - \left(\frac{\sin 27^\circ}{\cos 27^\circ} + \frac{\cos 27^\circ}{\sin 27^\circ} \right) \\ &= \frac{2}{\sin 18^\circ} - \frac{2}{\sin 54^\circ} \\ &= \left(\frac{2 \times 4}{\sqrt{5}-1} - \frac{2 \times 4}{\sqrt{5}+1} \right) \\ &= 8 \left(\frac{1}{\sqrt{5}-1} - \frac{1}{\sqrt{5}+1} \right) \\ &= 8 \left(\frac{\sqrt{5}+1}{(\sqrt{5}-1)(\sqrt{5}+1)} - \frac{\sqrt{5}-1}{(\sqrt{5}-1)(\sqrt{5}+1)} \right) \\ &= 8 \left(\frac{2}{5-1} \right) = 4 \end{aligned}$$

2. (4)

We need to find the value of

$$\begin{aligned} & 36(4 \cos^2 9^\circ - 1)(4 \cos^2 27^\circ - 1)(4 \cos^2 81^\circ - 1)(4 \cos^2 243^\circ - 1) \\ &= 36[(4 \cos^2 9^\circ - 1)(4 \sin^2 9^\circ - 1)(4 \cos^2 27^\circ - 1)(4 \sin^2 27^\circ - 1)] \\ &= 36[4(2 \cos 9^\circ \sin 9^\circ)^2 - 4 \cos^2 9^\circ - 4 \sin^2 9^\circ + 1] [4(2 \sin 27^\circ \cos 9^\circ)^2 - 4 \cos^2 27^\circ - 4 \sin^2 27^\circ + 1] \\ &= 36[(4 \sin^2 18^\circ - 4 + 1)(4 \sin^2 54^\circ - 4 + 1)] \\ &= 36[(4 \sin^2 18^\circ - 3)(4 \sin^2 54^\circ - 3)] \\ &= 36 \left[\left\{ \frac{(\sqrt{5}-1)^2}{4} - 3 \right\} \left\{ \frac{(\sqrt{5}+1)^2}{4} - 3 \right\} \right] \\ &= 36 \left[\left\{ \frac{(6-2\sqrt{5}-12)}{4} \right\} \left\{ \frac{(6+2\sqrt{5}-12)}{4} \right\} \right] \\ &= -36 \left[\left\{ \frac{(2\sqrt{5})^2 - 36}{16} \right\} \right] = 36 \end{aligned}$$

Hence this is the correct option.

3. (1)

Given,

$$\text{Expression } 96 \cdot \cos \frac{\pi}{33} \cdot \cos \frac{2\pi}{33} \cdot \cos \frac{4\pi}{33} \cdot \dots \dots \cos \frac{16\pi}{33}$$

Now we know that,

$$\cos A \cdot \cos 2A \cdot \cos 2^2 A \cdot \cos 2^3 A \cdot \dots \cdot \cos 2^{n-1} A = \frac{\sin 2^n A}{2^n \sin A}$$

Now using the above formula in given expression we get,

$$\begin{aligned} & 96 \cdot \cos \frac{\pi}{33} \cdot \cos \frac{2\pi}{33} \cdot \cos \frac{4\pi}{33} \cdot \dots \dots \cos \frac{16\pi}{33} \\ &= 96 \times \frac{\sin \frac{32\pi}{33}}{2^5 \sin \frac{\pi}{33}} \\ &= 96 \times \frac{\sin \left(\pi - \frac{\pi}{33} \right)}{2^5 \sin \frac{\pi}{33}} \\ &= 96 \times \frac{\sin \left(\frac{\pi}{33} \right)}{2^5 \sin \frac{\pi}{33}} \{ \text{as } \sin(\pi - \alpha) = \sin \alpha \} \\ &= 96 \times \frac{1}{32} = 3 \end{aligned}$$